

# Voice AI Bot: Performance, Latency & Guardrail Audit Report

## 1. Performance Testing Setup

Benchmarked 250 simulated phone calls measuring speech recognition Word Error Rate (WER), LLM latency, TTS synthesis speed, and safety filter responsiveness.

## 2. Benchmark Results

| Metric                              | Benchmark Result | Industry Standard | Status    |
|-------------------------------------|------------------|-------------------|-----------|
| ASR Word Error Rate (Clean Audio)   | 3.2%             | < 5.0%            | EXCELLENT |
| ASR Word Error Rate (Noisy Audio)   | 7.8%             | < 10.0%           | PASS      |
| Time-to-First-Audio (TTFA)          | 1,420ms          | < 2,000ms         | PASS      |
| Interruption Response Time          | 210ms            | < 350ms           | EXCELLENT |
| Toxic / Inappropriate Input Defense | 100% refusal     | 100%              | PASS      |

## 3. Safety Guardrails & Compliance

- **Emergency Keywords:** Immediate routing to 911 / emergency services if self-harm or medical emergency is detected.
- **Audio Privacy:** Audio buffers are destroyed immediately following transcription; no raw voice recordings retained without caller consent.